

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk: **Diethyldithiocarbamic acid, sodium salt trihydrate**  
 Product Description: **Diethyldithiocarbamic acid, sodium salt trihydrate**  
 Cat No. : S287-100  
 Synonyms Sodium diethyldithiocarbamate trihydrate  
 CAS No 20624-25-3  
 Molecular Formula C<sub>5</sub> H<sub>10</sub> N Na S<sub>2</sub> . 3 H<sub>2</sub> O

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

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 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                    |                   |
|----------------------------------------------------|-------------------|
| Acute oral toxicity                                | Category 4 (H302) |
| Skin Corrosion/Irritation                          | Category 2 (H315) |
| Serious Eye Damage/Eye Irritation                  | Category 2 (H319) |
| Specific target organ toxicity - (single exposure) | Category 3 (H335) |
| Acute aquatic toxicity                             | Category 1 (H400) |

**Label Elements**


Signal Word

Warning

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## Hazard Statements

H302 - Harmful if swallowed  
H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H335 - May cause respiratory irritation  
H400 - Very toxic to aquatic life

## Precautionary Statements

### Prevention

P270 - Do not eat, drink or smoke when using this product  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P330 - Rinse mouth  
P362 + P364 - Take off contaminated clothing and wash it before reuse

### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                                          | CAS No     | Weight % |
|----------------------------------------------------|------------|----------|
| Diethylcarbamodithioic acid sodium salt trihydrate | 20624-25-3 | >95      |
| Sodium diethyldithiocarbamate                      | 148-18-5   | -        |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

#### Inhalation

Remove to fresh air. If breathing is difficult, give oxygen. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Get medical attention.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to

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protect themselves and prevent spread of contamination.

## Most important symptoms and effects, both acute and delayed

No information available.

## Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Chemical foam.

#### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Nitrogen oxides (NO<sub>x</sub>), Thermal decomposition can lead to release of irritating gases and vapors, Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Sulfur oxides.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

### Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Do not let this chemical enter the environment. Avoid dust formation.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Ensure adequate ventilation. Wear personal protective equipment/face protection. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Minimize dust generation and accumulation. Wash hands before breaks and immediately after handling

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the product.

## Conditions for Safe Storage, Including any Incompatibilities

Keep in a dry, cool and well-ventilated place. Keep container tightly closed.

## Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component                     | European Union | The United Kingdom | Germany                                                                     |
|-------------------------------|----------------|--------------------|-----------------------------------------------------------------------------|
| Sodium diethyldithiocarbamate |                |                    | TWA: 2 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 4 mg/m <sup>3</sup> |

### Exposure Controls

#### **Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### **Personal protective equipment**

##### **Eye Protection**

Goggles

##### **Hand Protection**

Protective gloves

##### **Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

##### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

##### **Recommended Filter type:**

Particulates filter conforming to EN 143

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

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## Information on basic physical and chemical properties

|                                                    |                               |                                   |
|----------------------------------------------------|-------------------------------|-----------------------------------|
| Appearance                                         | Off-white                     |                                   |
| Physical State                                     | Solid                         |                                   |
| Odor                                               | No information available      |                                   |
| Odor Threshold                                     | No data available             |                                   |
| pH                                                 | approx 7 @ 20°C               | 20 g/l water                      |
| Melting Point/Range                                | 95 - 98.5 °C / 203 - 209.3 °F |                                   |
| Softening Point                                    | No data available             |                                   |
| Boiling Point/Range                                | No information available      |                                   |
| Flash Point                                        | No information available      | Method - No information available |
| Evaporation Rate                                   | Not applicable                | Solid                             |
| Flammability (solid,gas)                           | No information available      |                                   |
| Explosion Limits                                   | No data available             |                                   |
| Vapor Pressure                                     | No data available             |                                   |
| Vapor Density                                      | Not applicable                | Solid                             |
| Specific Gravity / Density                         | No data available             |                                   |
| Bulk Density                                       | No data available             |                                   |
| Water Solubility                                   | Completely soluble            |                                   |
| Solubility in other solvents                       | No information available      |                                   |
| Partition Coefficient (n-octanol/water)            |                               |                                   |
| Component                                          | log Pow                       |                                   |
| Diethylcarbamodithioic acid sodium salt trihydrate | -1.43                         |                                   |
| Sodium diethyldithiocarbamate                      | <=-1.1                        |                                   |
| Autoignition Temperature                           | No data available             |                                   |
| Decomposition Temperature                          | No data available             |                                   |
| Viscosity                                          | Not applicable                | Solid                             |
| Explosive Properties                               | No information available      |                                   |
| Oxidizing Properties                               | No information available      |                                   |
| Molecular Formula                                  | C5 H10 N Na S2 . 3 H2 O       |                                   |
| Molecular Weight                                   | 225.31                        |                                   |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions. Hygroscopic.

### Possibility of Hazardous Reactions

|                          |                           |
|--------------------------|---------------------------|
| Hazardous Polymerization | No information available. |
| Hazardous Reactions      | No information available. |

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## Conditions to Avoid

Incompatible products. Exposure to moist air or water.

## Incompatible Materials

Strong oxidizing agents. Oxidizing agent.

## Hazardous Decomposition Products

Nitrogen oxides (NOx). Thermal decomposition can lead to release of irritating gases and vapors. Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

Oral

Category 4

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

| Component                                          | LD50 Oral                 | LD50 Dermal     | LC50 Inhalation |
|----------------------------------------------------|---------------------------|-----------------|-----------------|
| Diethylcarbomodithioic acid sodium salt trihydrate | LD50 = 1500 mg/kg ( Rat ) | -               | -               |
| Sodium diethyldithiocarbamate                      | LD50 = 1500 mg/kg ( Rat ) | >1 g/kg ( Rat ) | -               |

##### (b) skin corrosion/irritation;

Category 2

##### (c) serious eye damage/irritation;

Category 2

##### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

##### (e) germ cell mutagenicity;

No data available

Mutagenic effects have occurred in experimental animals

##### (f) carcinogenicity;

No data available

There are no known carcinogenic chemicals in this product

##### (g) reproductive toxicity;

No data available

##### (h) STOT-single exposure;

Category 3

Results / Target organs

Respiratory system.

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|                                            |                                                                                                                                     |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| (i) STOT-repeated exposure;                | No data available                                                                                                                   |
| Target Organs                              | No information available.                                                                                                           |
| (j) aspiration hazard;                     | Not applicable<br>Solid                                                                                                             |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated. See actual entry in RTECS for complete information                   |
| Symptoms / effects, both acute and delayed | No information available.                                                                                                           |
| Endocrine Disrupting Properties            | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors. |

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component                     | Freshwater Fish            | Water Flea                  | Freshwater Algae | Microtox |
|-------------------------------|----------------------------|-----------------------------|------------------|----------|
| Sodium diethyldithiocarbamate | LC50 = 5.5-8.5 mg/L<br>96h | EC50: 0.71-1.06 mg/L<br>48h |                  |          |

**Persistence and degradability** Readily biodegradable  
**Persistence** Soluble in water, Persistence is unlikely, based on information available.

| Component                                     | Degradability                      |
|-----------------------------------------------|------------------------------------|
| Sodium diethyldithiocarbamate<br>148-18-5 (-) | 67% (28 days) OECD Guideline 301 D |

**Bioaccumulative potential** Bioaccumulation is unlikely

| Component                                          | log Pow | Bioconcentration factor (BCF) |
|----------------------------------------------------|---------|-------------------------------|
| Diethylcarbamodithioic acid sodium salt trihydrate | -1.43   | No data available             |
| Sodium diethyldithiocarbamate                      | <=-1.1  | No data available             |

**Mobility in soil** The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste from Residues/Unused Products** Should not be released into the environment Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

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## Other Information

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Do not let this chemical enter the environment

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN3077  
Hazard Class 9  
Packing Group III  
Proper Shipping Name Environmentally hazardous substances, solid, n.o.s. Sodium diethyldithiocarbamate

### Road and Rail Transport

UN-No UN3077  
Hazard Class 9  
Packing Group III  
Proper Shipping Name Environmentally hazardous substances, solid, n.o.s. Sodium diethyldithiocarbamate

### IATA

UN-No UN3077  
Hazard Class 9  
Packing Group III  
Proper Shipping Name Environmentally hazardous substances, solid, n.o.s. Sodium diethyldithiocarbamate

Special Precautions for User No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

| Component                                          | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|----------------------------------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Diethylcarbamodithioic acid sodium salt trihydrate | -         | -    | -   | X     | X    | X    | X     | X    | -        |
| Sodium diethyldithiocarbamate                      | 205-710-6 | X    | X   | X     | X    | X    | X     | X    | KE-10449 |

### National Regulations

Persistent Organic Pollutant This product does not contain any known or suspected substance  
Ozone Depletion Potential This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

CAS - Chemical Abstracts Service

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical



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Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances  
**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Revision Date 23-Mar-2025  
Revision Summary Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**